

# FREEDOM INTERNATIONAL SCHOOL

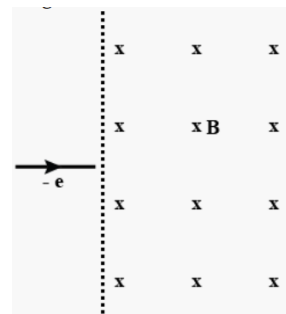
## WORKSHEET

### PHYSICS

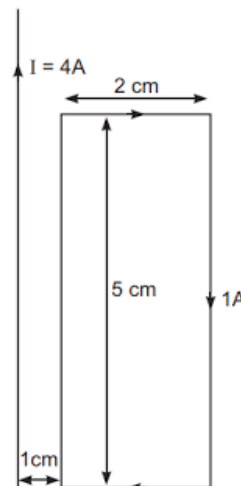
### CLASS XII

### MOVING CHARGES AND MAGNETISM

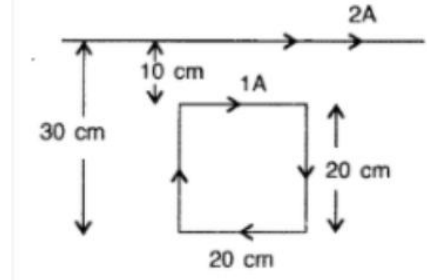
1. A solenoid of length 1.5 m, has a radius of 1.5 cm and has a total of 1500 turns wound on it. It carries a current of 3 A. Calculate the magnitude of the axial magnetic field inside the solenoid. If an electron had to move with a speed of  $2 \times 10^4$  m/s along the axis of this current carrying solenoid, what would the force experienced by this electron be?
2. An electron is moving at a speed of  $10^6$  m/s in a direction parallel to a current of 5 A, flowing through an infinitely long straight wire, separated by a perpendicular distance of 10 cm in air. Calculate the magnitude of the force experienced by the electron.
3. An electron moving horizontally with a velocity of  $4 \times 10^4$  m/s enters a region of uniform magnetic field of  $10^{-5}$  T acting as shown in the figure. Draw its trajectory and find out the time it takes to come out of the region of the magnetic field.



4. An electron after being accelerated through a potential difference of 100 V enters a uniform magnetic field of 0.004 T perpendicular to its direction of motion. Calculate the radius of the path described by the electron.
5. A wire AB is carrying a steady current of 12 A and is lying on the table. Another wire CD carrying 5 A is held directly above AB at a height of 1 mm. Find the mass per unit length of the wire CD so that it remains suspended at its position when left free. Give the direction of the current flowing in CD with respect to that in AB. [Take  $g = 10 \text{ m/s}^2$ ]
6. A rectangular loop of wire of size 2 cm x 5 cm carries a steady current of 1 A. A straight long wire carrying 4 A current is kept near the loop as shown in the figure. If the loop and the wire are coplanar, find
  - (i) the torque acting on the loop
  - (ii) the magnitude and direction of the force on the loop due to the current carrying wire.



7. A square loop of side 20 cm carrying current of 1 A is kept near an infinitely long straight wire carrying a current of 2 A in the same plane as shown in the figure. Calculate the magnitude and direction of the net force exerted on the loop due to the current carrying conductor.



8. A square shaped plane coil of area  $100 \text{ cm}^2$  of 200 turns carries a steady current of 5 A. It is placed in a uniform magnetic field of 0.2 T acting perpendicular to the plane of the coil. Calculate the torque on the coil when its plane makes an angle of  $60^\circ$  with the direction of the field. In which orientation will the coil be in stable equilibrium?
9. In a galvanometer there is a deflection of 10 divisions per mA. The internal resistance of the galvanometer is  $60 \Omega$ . If a shunt of  $2.5 \Omega$  is connected to the galvanometer and there are 50 divisions in all, on the scale of galvanometer, what maximum current can this galvanometer read?
10. A galvanometer can be converted into a voltmeter to measure up to:
- (i) V volt by connecting a resistance of  $2 \text{ k}\Omega$  in series with the galvanometer.
  - (ii) 2 V volt by connecting a resistance  $5 \text{ k}\Omega$  in series with the galvanometer.
- Calculate the resistance that should be connected in series with the galvanometer to convert it into a voltmeter to measure up to  $V/2$  volts.
11. A galvanometer of resistance  $16 \Omega$  shows full scale deflection for a current of 4 mA. How will you convert it into a voltmeter to measure a voltage up to 3 V?